

Console Variables Tester

Console variables are a fundamental part of game development in Unreal Engine 5, offering developers the ability to control various engine parameters on-the-fly. These variables can greatly influence the behavior and performance of a game. Let's explore the utility of Console Variables and the Console Variables Tester tool in Unreal Engine 5.

What are Console Variables?

Console variables in Unreal Engine 5 are variables that can be accessed and modified through the developer console. These variables control a wide variety of engine parameters, including graphics settings, debug options, AI behavior, physics properties, and more.

Console Variables Tester

In Unreal Engine 5, the Console Variables Tester is a powerful tool that assists developers in managing and testing the impact of these console variables on their game. This tool allows developers to quickly change the values of console variables and observe the effects on the game in real-time, without needing to restart or recompile the game. Here's a quick guide on how to use the Console Variables Tester:

- **Access the Console:** The first step is to open the console in the Unreal Editor. You can do this by pressing the tilde key (~) on your keyboard.
- **Find a Variable:** To find a specific console variable, you can use the find command followed by the name or part of the name of the variable you're looking for. This will return a list of all console variables matching your search term.
- **Modify a Variable:** Once you've found the console variable you want to modify, you can change its value directly from the console. Simply type the name of the variable followed by the new value and hit enter.
- **Observe the Effects:** After modifying a console variable, the changes will take effect immediately, and you can observe the impact on your game in real-time.

The Console Variables Tester is particularly useful for optimization and debugging. By adjusting different console variables, developers can identify performance bottlenecks, test the effects of different graphics settings, and more.

Using the Gameplay Debugger

Game development is a complex task, with countless variables and parameters at play at any given moment. To help developers manage this complexity, Unreal Engine 5 provides a suite of debugging tools, one of the most notable being the Gameplay Debugger. This invaluable tool can assist you in diagnosing and rectifying problems that can occur during the development of your game.

What is the Gameplay Debugger?

The Gameplay Debugger is a real-time, in-engine tool in Unreal

Engine 5 that allows developers to visualize and understand what's happening behind the scenes of their game. It provides live insights into the game's running state, such as artificial intelligence (AI) behavior, navigation paths, physics interactions,



and more. With the Gameplay Debugger, you can essentially 'see through the matrix' of your game, dissecting the behaviors and interactions that make up your virtual world.

How to use the Gameplay Debugger?

- **Enabling the Debugger:** First, you need to open the Gameplay Debugger. This can usually be done by pressing the apostrophe (') key while in the play mode within the editor.
- **Selecting a Debug Category:** Once the debugger is open, you'll see several categories of game parameters that you can debug. Use the number keys to toggle different debug categories on and off. Some common categories include AI, Navigation, Physics, and more.
- **Inspecting the Data:** With a debug category enabled, you'll see a visual representation of that category's data overlaid on your game. For example, if you've selected the AI category, you might see lines indicating AI sightlines or colored areas indicating detected sounds.
- **Adjusting the Settings:** The Gameplay Debugger allows you to customize what data is displayed and how it's presented. These settings can be accessed through the 'Gameplay Debugger' option in the 'Edit' menu of the Unreal Engine 5 editor. Here, you can add or remove debug categories, change display settings, and more.

The Gameplay Debugger is not just a tool for finding bugs. It's also an excellent way to understand how different parts of your game interact. By studying the live data from the debugger, you can gain insights into how your game works under the hood, which can help you make more informed design decisions and create more polished, well-optimized games.

Optimising rendering with PSO Caches

Pipeline State Objects (PSOs) represent the state of the graphics pipeline in Unreal Engine 5, and they can significantly influence